

Assignment 1

2020-06-28

- 1) Examine the effects of the persistence of core inflation on the model properties.
 - Create three parametrizations, which differ only in the value of the persistence in the core inflation Phillips curve. Set the values to 0.35, 0.5, and 0.65.
 - Create a model report containing the impulse responses only.
 - Create a historical filter report containing the trend/gap decompositions only.
 - Identify the differences between the three parametrizations. Which responses are significantly different? Which gaps are significantly different? Which parametrization gives the most reasonable results?
- 2) Examine the effects of the calibrated volatility of structural deficit.
 - Create three parametrizations, which differ only in the value of the standard deviation of the structural deficit shock. Set the values to 0.05, 0.30, and 1.80.
 - Create a model report containing the impulse responses only.
 - Create a historical filter report containing the equation decompositions only.
 - Identify the differences between the three parametrizations. Which responses are significantly different? Which estimated variables are significantly different? Which parametrization gives the most reasonable results?
- 3) Examine the role of domestic marginal cost in food inflation.
 - Create a new model file, in which you add
 - one new variable, called `rmc_food`
 - one new parameter, called `bf4`
 - one new equation, giving `rmc_food` as a linear combination of `l_y_gap` and the already existing foreign cost component
 - modify the current food inflation equation so that it has the new `rmc_food` variable in place of the current marginal cost variable
 - Create three parametrizations, which differ only in the newly added `bf4` parameter. Set the values to 0, 0.40, and 0.80.
 - Create a model report containing the impulse responses only.
 - Create a historical filter report containing the trend/gap decompositions only.

- Identify the differences between the three parametrizations. Which responses are significantly different? Which estimated variables are significantly different? Which parametrization gives the most reasonable results?
- (This is a bit harder) Create the historical filter report containing also the equation decompositions. You'll get warnings about some decompositions not being correct. Identify the source of the issue, and correct it. (You'll have to modify the `addDecompositions.m` file in the directory `+codes/+reporting`)

4) Check the weights of the demand components.

- Calculate the average shares of the demand components on nominal GDP over the recent history, after 2010. Make sure you pay attention to how we modify investment to account for government investment.
- Create a new parametrization, which differs from the baseline only in these shares.
- Create a model report containing the impulse responses only.
- Create a historical filter report containing the trend/gap and equation decompositions, and the shock series.
- Identify the differences between the two parametrizations. Which responses are significantly different? Which estimated variables and shocks are significantly different? Which parametrization gives the more reasonable results?
- Create also the historical forecasts with the trends fixed. Compare the results from the two parametrizations. Which variables differ the most? Try to give an explanation for the difference. Which parametrization gives the more reasonable results?